Supporting Material for Fixed Geometric Health Decisions After LiDAR Degeneracy

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Abstract

This supplement records the formal endpoint distinctions, exact configuration, geometry-screen history, existing sensitivity analyses, development repeatability and complete policy-event ledger supporting the controlled transfer study. It typesets retained evidence; it introduces no new native runs or threshold fitting. The independent final sampling unit is a geometry pair. There are 48 pairs, 192 native scene/backend runs, and 192 corridor policy-event rows; the latter two counts must not be interpreted as 192 independent geometries.

I. Evidence scope and reading map

The main manuscript contains the primary operational definitions, fixed-policy results and interpretation. The supporting CSV files retain full precision; printed tables round values only for readability. The primary exports are `research_paper/evidence/final_geometry_v2_results_20261012/`. The exploratory miss diagnosis is separately recorded in `final_failure_analysis_20261012/`. Its categories and example selection were produced after the primary results and are not preregistered hypotheses. All examples in the main paper use the existing lowest-seed selection rule. Source hashes and the exact timeline choices are retained in the adjacent `support/asset_manifest.json` and `support/timeline_selection.json`. No test-set threshold, revised label, selected example or new bootstrap was used to replace a primary result during this manuscript revision.

II. Formal endpoint arguments

These elementary arguments state what the measurements do and do not establish. They are explanatory proofs, not claims of new mathematical theorems.

A. Relative motion is invariant to a constant world-frame change

Let $T(t) \in SE(3)$ map the body frame to a world frame and let a constant $A \in SE(3)$ redefine that world frame. Then

$$(AT(t_a))^{-1}(AT(t_b)) = T(t_a)^{-1}A^{-1}AT(t_b) = T(t_a)^{-1}T(t_b). \tag{1}$$

Thus each relative transform, and the discrepancy between estimated and reference relative transforms in the same body-frame convention, is unchanged by a constant left world transform. This does not prove invariance to time-varying frame errors, incorrect extrinsics, scale changes or estimator resets. Reset-crossing windows are explicitly unavailable. Good relative motion also does not imply small accumulated global position error.

B. A fixed geometric Hessian does not bound arbitrary bias

Assume a linear measurement model $y = Jx^\star + b$ with $J \in \mathbb{R}^{m\times n}$ full column rank. The unique least-squares solution is

$$\widehat{x} = (J^\top J)^{-1}J^\top y = x^\star + (J^\top J)^{-1}J^\top b. \tag{2}$$

For any vector $v \in \mathbb{R}^n$, choose $b = Jv$. Then $\widehat{x} - x^\star = v$, while $J^\top J$ is unchanged. Therefore a score computed only from this Hessian cannot give an unconditional finite error bound when b is unrestricted. If instead $\|b\| \le B$, then $\|\widehat{x} - x^\star\| \le B/\sigma_{\min}(J)$; the missing assumption matters. This is a linear-model counterexample, not evidence that this particular bias caused any native-run failure, nor a proof that geometry has no predictive value.

C. An active state need not produce a new recovery announcement

For consecutive available decision ticks let $z_j \in \{0, 1\}$ denote confirmed health and $a_j = z_j(1 - z_{j-1})$ a transition announcement. If $z_j = 1$ before reference-defined onset and remains one throughout the recovered interval, then $a_j = 0$ throughout that interval. The state overlaps recovered motion, but no new post-onset announcement exists. This proves that a missed declaration alone cannot establish continuously unhealthy output. Missing or reset evidence clears confirmation; this argument applies to an uninterrupted available segment. The online decision sees only past diagnostic records. Reference onset is retrospective and is never an input to the state machine.

III. Exact timing, decisions and uncertainty

Primary motion windows have nominal non-overlapping one-second boundaries. The nearest pose within ± 0.05 s is selected once per boundary, with earlier pose on a tie; actual duration must be 0.9–1.1 s. A window passes when relative translation error is at most 0.20 m and rotation error at most 5°. Recovery is the first three consecutive passing post-exit windows by exit plus 20 s. Onset is the first actual window start, confirmation the third end. A later required failed/unavailable window ends the episode at its start; otherwise it ends at the deadline. One relapse is observed per backend. Completed non-recoveries remain distinct from failed execution.

Decisions use a 10 Hz grid, latest same-segment record no older than 0.2 s, and three distinct healthy source records on consecutive ticks. A newer invalid record cannot be skipped. The minimum accepted-row count is six. The scalar score uses a 3 m lever arm and fixed $N\,0.001$ normalization; these are measurement-geometry conventions rather than a calibrated posterior. The fixed Schur adaptation uses condition ratio 10 and pseudoinverse relative rank tolerance 10^{-12}. It does not implement DCReg's full estimator or mitigation. Scalar thresholds are 166.16366016039856 (FAST-LIO) and 123.80876977779258 (Point-LIO), selected only on development corridor seeds 14–45. The development selection ceiling is 10% conditional initial false-healthy events; sensitivity is maximized and ties favor the higher threshold.

Event proportions use Wilson 95% intervals. The paired interior-motion effect uses each scene's median error on the predeclared central corridor region, corridor minus matched control, then equal geometry-group weighting. Its percentile interval uses 10,000 geometry-pair bootstrap replicates within the four fixed groups, PCG seed 20260926 and linear quantiles. Policies on the same geometry and timestamps within a run are not independent samples. Detected-only median delays condition on successful declarations and must be read with the missed-event counts.

IV. Simulator and native identities

The simulator uses 60 s runs, a right-handed z-up frame, opaque rectangular surfaces, 0.8 m/s nominal cruise and prescribed smooth three-dimensional motion. LiDAR has 16 beams over $\pm 15^\circ$, 360 azimuth samples and 10 Hz scans. Range noise standard deviation is 0.01 m. IMU rate is 200 Hz with white noise 0.002 rad/s and 0.02 m/s^2; initial bias standard deviations are 0.003 rad/s and 0.03 m/s^2. Random walks are 0.0001 rad/s/$\sqrt{\text{s}}$ and 0.001 m/s^2/$\sqrt{\text{s}}$ respectively. Floor and ceiling are at $z = -1$ and 2 m. The pre-entry room spans $x = -12$ to 0 and the post-exit room $x = L$ to 50 m, with sampled half-width 4.5–7.5 m. The control adds partial positive-side baffles at $x = Li/(n+1)$, $n = \max(4, \lceil L/(2\,\text{m})\rceil - 1)$, spanning $y = 0.65w$ to w and $z = -1$ to 1.5 m while preserving the open route. The serialized paired IMU stream and prescribed trajectory match. Surface returns change with the baffles.

Each backend has 96 completed final runs. The identities below agree across all 96 manifests for that backend, checked against the retained receipt hashes. The JSON records upstream commits, binary hashes, configuration hashes and worktree-patch hashes. A commit identifier alone is insufficient for these instrumented builds. Historical configuration filenames include "development" or "bootstrap"; receipt hashes establish their actual use in the final runs.


```
{
  "fastlio": {
    "upstream_commit": "7cc4175de6f8ba2edf34bab02a42195b141027e9",
    "binary_sha256": "4da32e8bed7756de1e4d41883f58c91f4fc954fd6f2b6f4d9b052e3568697792",
    "configuration_sha256": "b1f937855f08c1ff2cba66606d43cd65877869404e18807c4185f385404d0a67",
    "worktree_diff_sha256": "1397219a8765c2bc8c49adaf7e94dcc1e6006788acac06467e28e660876e51a0"
  },
  "pointlio": {
    "upstream_commit": "4b86a469eb5572e70ed575af25b5f15dd06e8e3c",
    "binary_sha256": "80516da9cafba98262d56c17cc26ec4da76e214049d3af7776248b4c3a342b17",
    "configuration_sha256": "d23bd8799c3834ac0acc1d23476a0a0c0cd72b09f7b7c542b5bfbe9813a4c98b",
    "worktree_diff_sha256": "1270c63b553f660ade9cd787c0e377d1486ac4f49018650f400f2cf52eb3445c"
  }
}
```


A. FAST-LIO configuration, verbatim

```
# Development feasibility only; not a frozen scientific operating point.
common:
  lid_topic: /sim/points
  imu_topic: /sim/imu
  time_sync_en: false
  time_offset_lidar_to_imu: 0.0
preprocess:
  lidar_type: 2
  scan_line: 16
  scan_rate: 10
  timestamp_unit: 0
```

```
  blind: 0.2
mapping:
  acc_cov: 0.1
  gyr_cov: 0.1
  b_acc_cov: 0.0001
  b_gyr_cov: 0.0001
  fov_degree: 180.0
  det_range: 60.0
  extrinsic_est_en: false
  extrinsic_T: [0.0, 0.0, 0.0]
  extrinsic_R: [1.0, 0.0, 0.0, 0.0, 1.0, 0.0, 0.0, 0.0, 1.0]
publish:
  path_en: false
  scan_publish_en: false
  dense_publish_en: false
  scan_bodyframe_pub_en: false
pcd_save:
  pcd_save_en: false
  interval: -1
feature_extract_enable: false
point_filter_num: 4
max_iteration: 3
filter_size_surf: 0.5
filter_size_map: 0.5
cube_side_length: 1000.0
runtime_pos_log_enable: false
```

B. Point-LIO configuration, verbatim

```
# Development-only T15 configuration for the frozen seed-14 x=-6 m simulator.
# Units and topic fields were checked against Point-LIO's pinned Velodyne path.
common:
  lid_topic: /sim/points
  imu_topic: /sim/imu
  con_frame: false
  con_frame_num: 1
  cut_frame: false
  cut_frame_time_interval: 0.1
  time_diff_lidar_to_imu: 0.0

preprocess:
  lidar_type: 2              # Point-LIO VELO16 enum
  scan_line: 16
  timestamp_unit: 0          # simulator PointCloud2 `time` values are seconds
  blind: 0.2                 # match simulator/T12 minimum range

mapping:
  imu_en: true
  extrinsic_est_en: false    # collocated simulated LiDAR and IMU, identity transform
  imu_time_inte: 0.005       # 200 Hz simulator IMU
  lidar_time_inte: 0.1       # 10 Hz simulator LiDAR
  satu_acc: 30.0             # uncalibrated simulator stress range; no saturation claim
  satu_gyro: 35.0
  acc_norm: 9.81             # input accelerometer uses m/s^2
  lidar_meas_cov: 0.01
  acc_cov_output: 500.0
  gyr_cov_output: 1000.0
  b_acc_cov: 0.0001
  b_gyr_cov: 0.0001
  imu_meas_acc_cov: 0.1
  imu_meas_omg_cov: 0.1
  gyr_cov_input: 0.01
  acc_cov_input: 0.1
  plane_thr: 0.1
  match_s: 81
  ivox_grid_resolution: 2.0
  gravity: [0.0, 0.0, -9.81]
  gravity_init: [0.0, 0.0, -9.81]
  extrinsic_T: [0.0, 0.0, 0.0]
  extrinsic_R: [1.0, 0.0, 0.0, 0.0, 1.0, 0.0, 0.0, 0.0, 1.0]

odometry:
  publish_odometry_without_downsample: false
```

```
publish:
  path_en: false
  scan_publish_en: false
  scan_bodyframe_pub_en: false

pcd_save:
  pcd_save_en: false
  interval: -1

use_imu_as_input: 0
prop_at_freq_of_imu: true
check_satu: true
init_map_size: 10
space_down_sample: true
point_filter_num: 4
filter_size_surf: 0.5
filter_size_map: 0.5
ivox_nearby_type: 6
runtime_pos_log_enable: false
```

V. Geometry screening and retained workflow history

The original screen failed: 68 candidates yielded short/narrow 12, short/wide 10, long/narrow 0 and long/wide 12 eligible layouts, insufficient for the planned balanced final design. Its exact historical byte digest could not be reconciled because the original bytes are unavailable. The retained reconciliation explicitly records `UNRESOLVED_ORIGINAL_BYTES_NOT_AVAILABLE`; this limitation is not silently converted into a successful provenance claim.

The revised, separately frozen geometry-only screen checked 64 candidates and selected 12 per group, 48 total, without inspecting final estimator accuracy or scores. Short lengths span 8–9 m, long 15–18 m; narrow half-widths 1.3–1.5 m, wide 2.5–3.0 m. Development lengths were 9–13 m and half-widths 1.6–2.4 m. Eligibility requires axial-normal fraction $f_c \leq 0.10$ in the corridor, $f_u \geq 0.15$ in the matched control, and $f_{+5} \geq 0.20$ at five seconds past exit. These fractions screen geometry, not observed recovery. The test population is conditional on this screen; it is not every possible corridor. Table rows below include unsuccessful and unused candidates.

TABLE I: All 64 revised-screen candidates. Length and half-width are in metres; fractions are rounded for display only.

Group	Seed	L	w	f_c	f_u	f_{+5}	Eligible	Used
SN	10000	8.612	1.462	0.053	0.203	0.255	yes	yes
SN	10001	8.239	1.465	0.051	0.207	0.280	yes	yes
SN	10002	8.827	1.355	0.040	0.194	0.296	yes	yes
SN	10003	8.288	1.444	0.051	0.205	0.271	yes	yes
SN	10004	8.333	1.348	0.046	0.201	0.277	yes	yes
SN	10005	8.011	1.347	0.046	0.199	0.291	yes	yes
SN	10006	8.396	1.332	0.047	0.200	0.271	yes	yes
SN	10007	8.665	1.449	0.049	0.203	0.274	yes	yes
SN	10008	8.095	1.328	0.045	0.199	0.287	yes	yes
SN	10009	8.212	1.469	0.048	0.202	0.290	yes	yes
SN	10010	8.462	1.366	0.047	0.201	0.279	yes	yes
SN	10011	8.285	1.447	0.045	0.198	0.295	yes	yes
SW	10100	8.097	2.642	0.099	0.241	0.225	yes	yes
SW	10101	8.953	2.698	0.092	0.234	0.201	yes	yes
SW	10102	8.112	2.918	0.104	0.240	0.194	no	no
SW	10103	8.687	2.999	0.102	0.239	0.217	no	no
SW	10104	8.639	2.508	0.090	0.235	0.235	yes	yes
SW	10105	8.482	2.755	0.098	0.238	0.223	yes	yes
SW	10106	8.191	2.597	0.095	0.238	0.204	yes	yes
SW	10107	8.248	2.533	0.094	0.239	0.210	yes	yes
SW	10108	8.246	2.900	0.099	0.236	0.190	no	no
SW	10109	8.485	2.835	0.093	0.233	0.182	no	no
SW	10110	8.492	2.544	0.096	0.239	0.215	yes	yes
SW	10111	8.389	2.514	0.094	0.238	0.232	yes	yes
SW	10112	8.896	2.740	0.093	0.236	0.224	yes	yes
SW	10113	8.440	2.822	0.101	0.238	0.224	no	no
SW	10114	8.299	2.968	0.109	0.245	0.211	no	no
SW	10115	8.083	2.798	0.105	0.243	0.218	no	no
SW	10116	8.838	2.524	0.087	0.230	0.203	yes	yes
SW	10117	8.158	2.595	0.098	0.241	0.224	yes	yes
SW	10118	8.199	2.628	0.099	0.240	0.228	yes	yes
LN	10200	17.639	1.317	0.021	0.186	0.284	yes	yes

Continued on next page

TABLE I – continued

Group	Seed	L	w	f_c	f_u	f_{+5}	Eligible	Used
LN	10201	15.938	1.366	0.024	0.136	0.273	no	no
LN	10202	17.228	1.371	0.022	0.193	0.284	yes	yes
LN	10203	16.566	1.303	0.022	0.194	0.294	yes	yes
LN	10204	17.895	1.308	0.020	0.182	0.293	yes	yes
LN	10205	17.871	1.352	0.022	0.186	0.262	yes	yes
LN	10206	16.123	1.486	0.027	0.210	0.269	yes	yes
LN	10207	16.510	1.498	0.027	0.208	0.251	yes	yes
LN	10208	16.061	1.372	0.024	0.202	0.279	yes	yes
LN	10209	16.762	1.412	0.024	0.200	0.253	yes	yes
LN	10210	16.260	1.353	0.024	0.202	0.261	yes	yes
LN	10211	15.277	1.465	0.029	0.149	0.256	no	no
LN	10212	16.456	1.355	0.023	0.200	0.267	yes	yes
LN	10213	17.091	1.356	0.022	0.193	0.292	yes	yes
LW	10300	16.311	2.507	0.049	0.246	0.209	yes	yes
LW	10301	15.125	2.975	0.059	0.227	0.185	no	no
LW	10302	16.561	2.689	0.051	0.250	0.184	no	no
LW	10303	15.435	2.600	0.053	0.213	0.220	yes	yes
LW	10304	15.587	2.703	0.056	0.215	0.210	yes	yes
LW	10305	15.215	2.560	0.053	0.213	0.206	yes	yes
LW	10306	15.284	2.876	0.059	0.223	0.204	yes	yes
LW	10307	17.092	2.790	0.053	0.245	0.212	yes	yes
LW	10308	15.146	2.816	0.058	0.224	0.199	no	no
LW	10309	15.736	2.605	0.052	0.210	0.199	no	no
LW	10310	15.769	2.792	0.056	0.214	0.218	yes	yes
LW	10311	16.215	2.750	0.054	0.253	0.221	yes	yes
LW	10312	17.333	2.889	0.052	0.243	0.216	yes	yes
LW	10313	17.324	2.980	0.052	0.246	0.175	no	no
LW	10314	15.317	2.606	0.053	0.215	0.193	no	no
LW	10315	16.027	2.858	0.057	0.259	0.205	yes	yes
LW	10316	17.468	2.929	0.054	0.245	0.206	yes	yes
LW	10317	16.050	2.799	0.053	0.256	0.183	no	no
LW	10318	16.590	2.547	0.049	0.245	0.224	yes	yes

All 192 selected native runs completed; none was replaced by a selective retry. A Point-LIO preflight failed for missing scratch storage before any replay. An earlier primary-analyzer invocation rejected extra shared-threshold metadata before loading poses; the retained compatibility adapter admits that metadata without altering the scientific analysis. Those workflow failures remain recorded in the evidence package and do not represent completed native non-recoveries. Earlier GEODE failures and Hilti-Oxford development-only runs do not enter these final simulated estimates.

VI. Additional analyses from the accepted exports

A. Threshold-free discrimination

The following scores describe ranking, not the performance of a replacement threshold. Tick weights give each geometry equal contribution; bootstrap units remain geometry clusters. Scalar scores increase with health; the directional score is the reciprocal maximum condition ratio. The truth target is membership in the reference recovered interval. All four estimates use 10,000 retained bootstrap replicates, with no undefined replicates. High AUC does not imply high declaration sensitivity at the frozen operating point.

TABLE II
Existing discrimination estimates. AP denotes average precision.

Backend	Policy	ROC-AUC [95% CI]	AP [95% CI]
FAST-LIO	Scalar	0.921 [0.865, 0.955]	0.985 [0.976, 0.991]
FAST-LIO	Fixed Schur	0.903 [0.839, 0.962]	0.984 [0.966, 0.996]
Point-LIO	Scalar	0.989 [0.982, 0.994]	0.990 [0.983, 0.995]
Point-LIO	Fixed Schur	0.900 [0.845, 0.945]	0.943 [0.900, 0.971]

B. Reference-label tolerance sensitivity

Half tolerance is 0.10 m and 2.5°; primary is 0.20 m and 5°; double is 0.40 m and 10°. Each still requires three consecutive one-second windows. These alternative reference labels do not replace the primary outcome or select a preferred threshold. They do not report retuned policy performance.

TABLE III
RECOVERED EVENTS UNDER EXISTING TOLERANCE SENSITIVITY LABELS.

Backend	Tolerance	Recovered	Rate (%)	Wilson 95%
FAST-LIO	Half	44/48	91.7	[80.4, 96.7]
FAST-LIO	Primary	46/48	95.8	[86.0, 98.8]
FAST-LIO	Double	47/48	97.9	[89.1, 99.6]
Point-LIO	Half	31/48	64.6	[50.4, 76.6]
Point-LIO	Primary	32/48	66.7	[52.5, 78.3]
Point-LIO	Double	33/48	68.8	[54.7, 80.1]

TABLE IV
TOLERANCE SENSITIVITY SEPARATED BY GEOMETRY GROUP.

Backend	Geometry	Tolerance	Recovered
FAST-LIO	Short/narrow	Half	11/12
FAST-LIO	Short/narrow	Primary	12/12
FAST-LIO	Short/narrow	Double	12/12
FAST-LIO	Short/wide	Half	10/12
FAST-LIO	Short/wide	Primary	10/12
FAST-LIO	Short/wide	Double	11/12
FAST-LIO	Long/narrow	Half	12/12
FAST-LIO	Long/narrow	Primary	12/12
FAST-LIO	Long/narrow	Double	12/12
FAST-LIO	Long/wide	Half	11/12
FAST-LIO	Long/wide	Primary	12/12
FAST-LIO	Long/wide	Double	12/12
Point-LIO	Short/narrow	Half	12/12
Point-LIO	Short/narrow	Primary	12/12
Point-LIO	Short/narrow	Double	12/12
Point-LIO	Short/wide	Half	3/12
Point-LIO	Short/wide	Primary	4/12
Point-LIO	Short/wide	Double	4/12
Point-LIO	Long/narrow	Half	12/12
Point-LIO	Long/narrow	Primary	12/12
Point-LIO	Long/narrow	Double	12/12
Point-LIO	Long/wide	Half	4/12
Point-LIO	Long/wide	Primary	4/12
Point-LIO	Long/wide	Double	5/12

C. Separate three-second motion-rate sensitivity

This analysis uses three-second relative motion windows starting every 0.1 s, not the primary three consecutive one-second criterion. It applies the existing translation/rotation rate cutoffs of 0.20 m/s and 5°/s. Each scene has 170 planned post-exit windows per geometry, all valid here. The overlapping windows are not independent observations. Tables show an equal-event mean pass fraction, rather than treating 8,160 windows as independent recovered events.

TABLE V
EXISTING THREE-SECOND RATE SENSITIVITY, POOLED ACROSS GEOMETRIES.

Backend	Scene	Events	Valid/planned	Mean pass (%)
FAST-LIO	Corridor	48	8160/8160	92.32
FAST-LIO	Control	48	8160/8160	94.61
Point-LIO	Corridor	48	8160/8160	64.15
Point-LIO	Control	48	8160/8160	81.67

TABLE VI
EXISTING THREE-SECOND RATE SENSITIVITY BY GROUP.

Backend	Geometry	Scene	Events	Valid/planned	Pass (%)
FAST-LIO	Short/narrow	corridor	12	2040/2040	90.98
FAST-LIO	Short/narrow	control	12	2040/2040	99.12
FAST-LIO	Short/wide	corridor	12	2040/2040	81.81
FAST-LIO	Short/wide	control	12	2040/2040	80.00
FAST-LIO	Long/narrow	corridor	12	2040/2040	100.00
FAST-LIO	Long/narrow	control	12	2040/2040	100.00
FAST-LIO	Long/wide	corridor	12	2040/2040	96.47
FAST-LIO	Long/wide	control	12	2040/2040	99.31
Point-LIO	Short/narrow	corridor	12	2040/2040	97.40
Point-LIO	Short/narrow	control	12	2040/2040	97.25
Point-LIO	Short/wide	corridor	12	2040/2040	26.08
Point-LIO	Short/wide	control	12	2040/2040	69.95
Point-LIO	Long/narrow	corridor	12	2040/2040	100.00
Point-LIO	Long/narrow	control	12	2040/2040	100.00
Point-LIO	Long/wide	corridor	12	2040/2040	33.14
Point-LIO	Long/wide	control	12	2040/2040	59.46

D. Development repeatability and its limits

The retained development audit compares four repeat runs (seeds 14 and 45, corridor/control) against their primary FAST-LIO runs. They add zero independent geometries and do not replace primary runs. Some native outputs differ; repeatability is not a claim of universally byte-identical native execution. Table VII reports absolute changes in aligned one-second translation error and counts of changed confirmed-health ticks across the run. The seed-14 corridor comparison has matching measurement hashes and identical error outputs; other comparisons do not all have that property. Corridor event decisions agree in this audit despite differing timelines in seed 45. This limited development audit cannot certify final-population deterministic native replay.

TABLE VII
DEVELOPMENT REPEAT COMPARISONS; CHANGED TICK COUNTS ARE DESCRIPTIVE.

Seed	Scene	Median change (m)	Maximum (m)	Scalar ticks	Schur ticks
14	corridor	0.0000	0.0000	0	0
14	control	0.0097	0.1159	52	75
45	corridor	0.0057	0.1684	0	112
45	control	0.0094	0.4160	0	97

VII. Complete final policy-event ledger

Figure 1 shows every geometry and both policies. The tables then list all 192 policy-event rows: 96 per backend, two policies sharing each of 48 corridor runs. There are also 96 control runs, used for the matched comparison. In the tables S denotes scalar and D the adapted directional rule; R means reference recovered and NR completed non-recovery. FH denotes initial false-healthy. A missing delay is not a zero delay. The mutually exclusive miss classes are "raw" (no raw healthy tick in the recovered interval), "confirm" (raw health without the required confirmation), and "active" (already-active confirmed health without a new post-onset transition). A dash means no missed recovery to classify, including non-recovery events. These classes explain decision behavior; they do not establish physical causation.

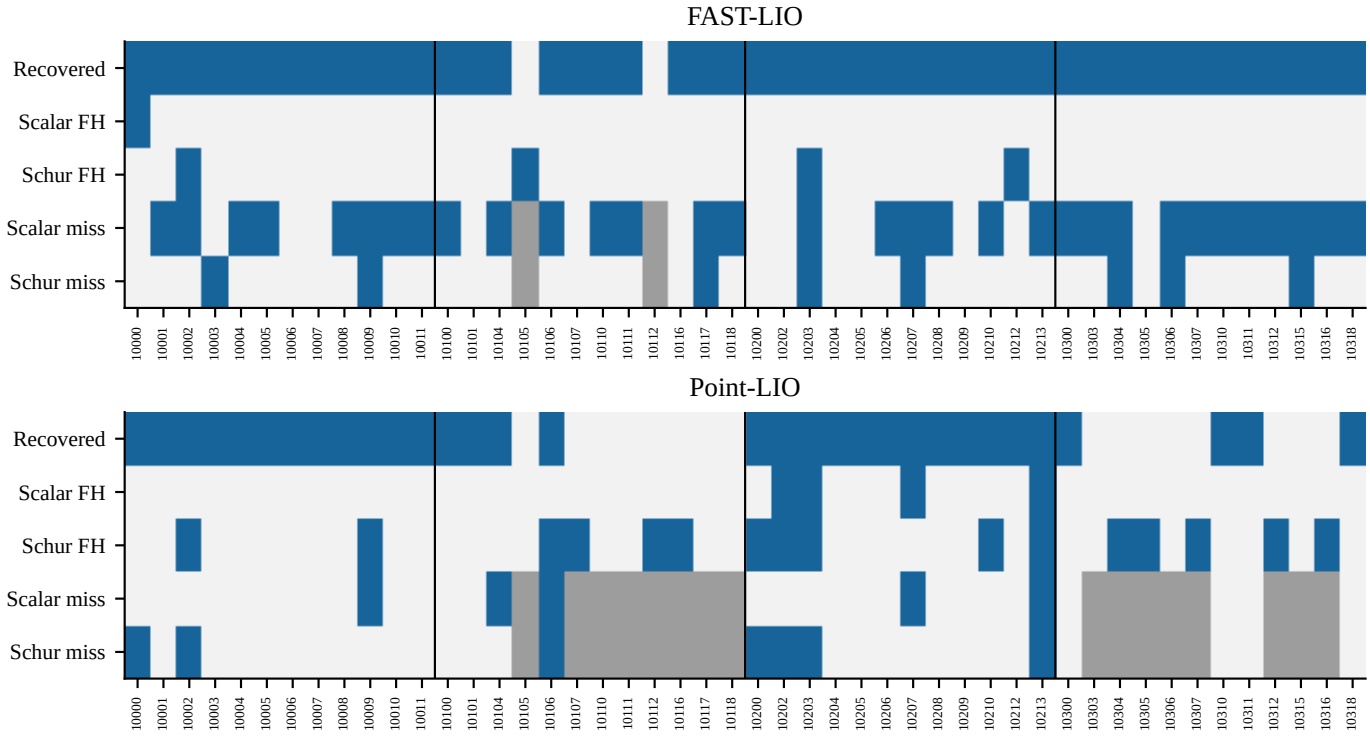


Fig. 1. Complete event map from retained exports. Columns are the 48 selected seeds, ordered within SN, SW, LN and LW groups. Dark blue marks a recovered event or a present FH/missed-declaration event in its named row. Light cells mean absence; grey cells are not applicable because reference recovery did not occur. Policy-event rows are paired within geometry, not independent replicates.

TABLE VIII: FAST-LIO: all 96 policy-event rows (48 shared geometries; S = scalar, D = adapted directional rule).

Group	Seed	Policy	Truth	FH	Declared	Delay (s)	Miss class
SN	10000	D	R	no	yes	13.400	–
SN	10000	S	R	yes	yes	15.400	–
SN	10001	D	R	no	yes	13.600	–
SN	10001	S	R	no	no	–	raw
SN	10002	D	R	yes	yes	12.900	–
SN	10002	S	R	no	no	–	raw
SN	10003	D	R	no	no	–	raw
SN	10003	S	R	no	yes	5.500	–
SN	10004	D	R	no	yes	12.300	–
SN	10004	S	R	no	no	–	raw
SN	10005	D	R	no	yes	5.800	–
SN	10005	S	R	no	no	–	raw
SN	10006	D	R	no	yes	13.300	–
SN	10006	S	R	no	yes	0.500	–
SN	10007	D	R	no	yes	6.000	–
SN	10007	S	R	no	yes	0.100	–
SN	10008	D	R	no	yes	8.500	–
SN	10008	S	R	no	no	–	raw
SN	10009	D	R	no	no	–	raw
SN	10009	S	R	no	no	–	raw

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TABLE VIII – continued

Group	Seed	Policy	Truth	FH	Declared	Delay (s)	Miss class
SN	10010	D	R	no	yes	10.200	–
SN	10010	S	R	no	no	–	confirm
SN	10011	D	R	no	yes	0.500	–
SN	10011	S	R	no	no	–	raw
SW	10100	D	R	no	yes	12.100	–
SW	10100	S	R	no	no	–	raw
SW	10101	D	R	no	yes	9.300	–
SW	10101	S	R	no	yes	0.900	–
SW	10104	D	R	no	yes	6.100	–
SW	10104	S	R	no	no	–	raw
SW	10105	D	NR	yes	no	–	–
SW	10105	S	NR	no	no	–	–
SW	10106	D	R	no	yes	16.400	–
SW	10106	S	R	no	no	–	raw
SW	10107	D	R	no	yes	13.800	–
SW	10107	S	R	no	yes	0.400	–
SW	10110	D	R	no	yes	10.200	–
SW	10110	S	R	no	no	–	raw
SW	10111	D	R	no	yes	9.500	–
SW	10111	S	R	no	no	–	raw
SW	10112	D	NR	no	no	–	–
SW	10112	S	NR	no	no	–	–
SW	10116	D	R	no	yes	13.900	–
SW	10116	S	R	no	yes	0.200	–
SW	10117	D	R	no	no	–	confirm
SW	10117	S	R	no	no	–	raw
SW	10118	D	R	no	yes	4.300	–
SW	10118	S	R	no	no	–	raw
LN	10200	D	R	no	yes	1.200	–
LN	10200	S	R	no	yes	5.700	–
LN	10202	D	R	no	yes	1.100	–
LN	10202	S	R	no	yes	5.400	–
LN	10203	D	R	yes	no	–	active
LN	10203	S	R	no	no	–	raw
LN	10204	D	R	no	yes	0.900	–
LN	10204	S	R	no	yes	1.800	–
LN	10205	D	R	no	yes	0.200	–
LN	10205	S	R	no	yes	1.500	–
LN	10206	D	R	no	yes	7.500	–
LN	10206	S	R	no	no	–	raw
LN	10207	D	R	no	no	–	confirm
LN	10207	S	R	no	no	–	raw
LN	10208	D	R	no	yes	3.400	–
LN	10208	S	R	no	no	–	raw
LN	10209	D	R	no	yes	1.300	–
LN	10209	S	R	no	yes	0.800	–
LN	10210	D	R	no	yes	12.600	–
LN	10210	S	R	no	no	–	confirm
LN	10212	D	R	yes	yes	17.800	–
LN	10212	S	R	no	yes	0.300	–
LN	10213	D	R	no	yes	0.800	–
LN	10213	S	R	no	no	–	raw
LW	10300	D	R	no	yes	13.400	–
LW	10300	S	R	no	no	–	raw
LW	10303	D	R	no	yes	8.900	–
LW	10303	S	R	no	no	–	raw
LW	10304	D	R	no	no	–	raw
LW	10304	S	R	no	no	–	raw
LW	10305	D	R	no	yes	2.400	–
LW	10305	S	R	no	yes	0.000	–
LW	10306	D	R	no	no	–	raw
LW	10306	S	R	no	no	–	raw
LW	10307	D	R	no	yes	4.500	–
LW	10307	S	R	no	no	–	raw
LW	10310	D	R	no	yes	1.800	–
LW	10310	S	R	no	no	–	raw
LW	10311	D	R	no	yes	5.000	–
LW	10311	S	R	no	no	–	raw
LW	10312	D	R	no	yes	4.000	–
LW	10312	S	R	no	no	–	raw

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TABLE VIII – continued

Group	Seed	Policy	Truth	FH	Declared	Delay (s)	Miss class
LW	10315	D	R	no	no	–	raw
LW	10315	S	R	no	no	–	raw
LW	10316	D	R	no	yes	3.400	–
LW	10316	S	R	no	no	–	raw
LW	10318	D	R	no	yes	5.400	–
LW	10318	S	R	no	no	–	raw

TABLE IX: Point-LIO: all 96 policy-event rows (48 shared geometries; S = scalar, D = adapted directional rule).

Group	Seed	Policy	Truth	FH	Declared	Delay (s)	Miss class
SN	10000	D	R	no	no	–	raw
SN	10000	S	R	no	yes	5.000	–
SN	10001	D	R	no	yes	3.400	–
SN	10001	S	R	no	yes	4.900	–
SN	10002	D	R	yes	no	–	active
SN	10002	S	R	no	yes	4.300	–
SN	10003	D	R	no	yes	11.800	–
SN	10003	S	R	no	yes	7.100	–
SN	10004	D	R	no	yes	3.300	–
SN	10004	S	R	no	yes	0.500	–
SN	10005	D	R	no	yes	3.200	–
SN	10005	S	R	no	yes	4.500	–
SN	10006	D	R	no	yes	11.700	–
SN	10006	S	R	no	yes	4.100	–
SN	10007	D	R	no	yes	0.000	–
SN	10007	S	R	no	yes	2.700	–
SN	10008	D	R	no	yes	7.800	–
SN	10008	S	R	no	yes	4.900	–
SN	10009	D	R	yes	yes	0.100	–
SN	10009	S	R	no	no	–	raw
SN	10010	D	R	no	yes	2.200	–
SN	10010	S	R	no	yes	3.400	–
SN	10011	D	R	no	yes	0.000	–
SN	10011	S	R	no	yes	2.700	–
SW	10100	D	R	no	yes	3.200	–
SW	10100	S	R	no	yes	0.200	–
SW	10101	D	R	no	yes	10.400	–
SW	10101	S	R	no	yes	0.300	–
SW	10104	D	R	no	yes	1.200	–
SW	10104	S	R	no	no	–	raw
SW	10105	D	NR	no	no	–	–
SW	10105	S	NR	no	no	–	–
SW	10106	D	R	yes	no	–	raw
SW	10106	S	R	no	no	–	raw
SW	10107	D	NR	yes	no	–	–
SW	10107	S	NR	no	no	–	–
SW	10110	D	NR	no	no	–	–
SW	10110	S	NR	no	no	–	–
SW	10111	D	NR	no	no	–	–
SW	10111	S	NR	no	no	–	–
SW	10112	D	NR	yes	no	–	–
SW	10112	S	NR	no	no	–	–
SW	10116	D	NR	yes	no	–	–
SW	10116	S	NR	no	no	–	–
SW	10117	D	NR	no	no	–	–
SW	10117	S	NR	no	no	–	–
SW	10118	D	NR	no	no	–	–
SW	10118	S	NR	no	no	–	–
LN	10200	D	R	yes	no	–	active
LN	10200	S	R	no	yes	2.800	–
LN	10202	D	R	yes	no	–	active
LN	10202	S	R	yes	yes	14.400	–
LN	10203	D	R	yes	no	–	active
LN	10203	S	R	yes	yes	18.100	–
LN	10204	D	R	no	yes	0.700	–
LN	10204	S	R	no	yes	4.700	–

Continued on next page

TABLE IX – continued

Group	Seed	Policy	Truth	FH	Declared	Delay (s)	Miss class
LN	10205	D	R	no	yes	10.700	–
LN	10205	S	R	no	yes	0.400	–
LN	10206	D	R	no	yes	0.500	–
LN	10206	S	R	no	yes	0.800	–
LN	10207	D	R	no	yes	7.900	–
LN	10207	S	R	yes	no	–	active
LN	10208	D	R	no	yes	3.000	–
LN	10208	S	R	no	yes	0.300	–
LN	10209	D	R	no	yes	9.900	–
LN	10209	S	R	no	yes	0.100	–
LN	10210	D	R	yes	yes	0.300	–
LN	10210	S	R	no	yes	0.300	–
LN	10212	D	R	no	yes	11.500	–
LN	10212	S	R	no	yes	2.200	–
LN	10213	D	R	yes	no	–	active
LN	10213	S	R	yes	no	–	active
LW	10300	D	R	no	yes	2.700	–
LW	10300	S	R	no	yes	8.200	–
LW	10303	D	NR	no	no	–	–
LW	10303	S	NR	no	no	–	–
LW	10304	D	NR	yes	no	–	–
LW	10304	S	NR	no	no	–	–
LW	10305	D	NR	yes	no	–	–
LW	10305	S	NR	no	no	–	–
LW	10306	D	NR	no	no	–	–
LW	10306	S	NR	no	no	–	–
LW	10307	D	NR	yes	no	–	–
LW	10307	S	NR	no	no	–	–
LW	10310	D	R	no	yes	0.300	–
LW	10310	S	R	no	yes	4.200	–
LW	10311	D	R	no	yes	0.100	–
LW	10311	S	R	no	yes	2.500	–
LW	10312	D	NR	yes	no	–	–
LW	10312	S	NR	no	no	–	–
LW	10315	D	NR	no	no	–	–
LW	10315	S	NR	no	no	–	–
LW	10316	D	NR	yes	no	–	–
LW	10316	S	NR	no	no	–	–
LW	10318	D	R	no	yes	0.000	–
LW	10318	S	R	no	yes	15.200	–

VIII. Machine-readable supporting files and reproduction

The adjacent `support/` folder includes the full-precision diagnostic-event ledger, exploratory diagnosis, all 64 screen candidates, all 192 native-run receipts, native build identities, copied configurations, selected timeline metadata and asset/source hashes. The full native ledger records backend, seed, group, scene, run identity, completion status, exit code, elapsed time, manifest digest and retained manifest path. Absolute original paths are provenance; the portable reproduction package supplies a relocation map.

The versioned reproduction package is listed below with its verified repository commit.

```
reproduction/lidar_recovery_20261023/
d696dab58d93bcd640e3651965246bf9158d2d0e
```

Its offline reanalysis regenerated 96 recovery labels, 192 event rows and 20 figures; discrete labels and decisions agreed exactly and maximum numerical difference was 7.1×10^{-15}. The original 197-entry implementation freeze remains separately verified. That check certifies the packaged offline reanalysis, not a complete native estimator replay on another machine. Current source and original frozen source are distinguished in the package.

The main manuscript and this supplement are drafts awaiting human author review, verified affiliations and declarations, venue selection and its exact submission requirements. Scientific review must assess comparator fidelity and whether this bounded empirical contribution is sufficient for the chosen venue. These materials support that review without promising publication acceptance.